

Convolutions: The Core Operation

Scaler School of Technology | Deep Learning I: Neural Networks | Week 4 — Session 1/2 | 27 May 2026



What Is This Lecture About?

This session introduces the **convolution operation** — the building block behind every image recognition system. You'll learn why fully-connected layers fail on images, and how a small sliding filter can detect patterns efficiently with far fewer parameters.

1 From FC to Local Connectivity

Why connecting every pixel to every neuron is wasteful. Images have spatial structure — nearby pixels are related, distant ones aren't. *Like reading a book page-by-page instead of all pages at once.*

2 The Convolution Operation

A small filter slides across the image, computing dot products at each position. You'll see how simple filters detect edges, corners, and textures. *Like a magnifying glass scanning across a photo — same lens, different locations.*

3 Padding, Stride, Parameter Sharing & Pooling

How to control output dimensions (padding, stride), why the same filter is reused everywhere (parameter sharing), and how pooling summarizes feature maps. *One formula governs it all: $(W - F + 2P)/S + 1$.*

What to Expect in the Session

- Live calculation: parameter counts (FC vs Conv)
- Step-by-step convolution walkthrough on a 5×5 grid
- ★ **Notebook demo:** applying edge detection filters to real images
- iPad drawings of filter sliding and receptive field growth
- Output size formula practice with quizzes
- Building a simple CNN and tracing shapes through each layer

How to Prepare (Optional but Helpful)

- ✓ **Recall** the dot product — multiply elementwise, then sum
- ✓ **Brush up on** matrix dimensions and how they propagate through layers (Week 1–2 material)
- ✓ **Think about:** why does a 224×224 image need different treatment than a 100-feature tabular dataset?
- ✓ **Optional skim:** 3Blue1Brown's "But what is a convolution?" video (first 5 min)

Duration: ~2 hours

Prerequisites: MLPs, forward pass, matrix operations

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